

Year 7 • Chapter 11 • Probability: answers (teacher copy)

11.01 Sample space (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: {H, T}
- Ex 1b: {1, 2, 3, 4, 5, 6}
- Ex 2a: {pass, fail}
- Ex 2b: {G, PG, M, MA15+, R18+}
- Ex 3a: 5; no, the sectors are different sizes
- Ex 3b: {Katrina, Biljana}

Easy (drill)

- | | | |
|-------|-------------------------|--------------------------|
| 1. 5 | 8. 12 | 15. 1, 2, 3, 4, 5, 6 |
| 2. 5 | 9. 2 | 16. 2, 3, 5, 7, 11 |
| 3. 52 | 10. 10 | 17. 1, 3, 5, 7, 9 |
| 4. 26 | 11. 4 | 18. hearts,
diamonds, |
| 5. 7 | 12. 7 | clubs, spades |
| 6. 8 | 13. B, O, K | 19. H, T |
| 7. 4 | 14. red, blue,
green | 20. C, A, T |

Medium (drill)

- | | | |
|-------|--------|---------|
| 1. 4 | 7. 8 | 13. no |
| 2. 84 | 8. 16 | 14. yes |
| 3. 18 | 9. 30 | 15. no |
| 4. 14 | 10. 12 | 16. yes |
| 5. 8 | 11. 12 | |
| 6. 8 | 12. 36 | |

Challenging

- {HH, HT, TH, TT}
- A-H, A-T, B-H, B-T, C-H, C-T, D-H, D-T
- 36 (6×6)

Extension

- 16
- 9

Exit ticket

- Level 1: {red, yellow, blue}
- Level 2: 6
- Level 3: no: it depends on the student

11.02 Probability (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: $\frac{1}{6}$
- Ex 1b: $\frac{1}{2}$
- Ex 2a: $\frac{2}{3}$
- Ex 2b: $\frac{3}{10}$ (C)
- Ex 3a: 20.2%
- Ex 3b: 86.1%

Easy (drill)

- | | | |
|-------------------|-------------------|-------------------|
| 1. $\frac{7}{10}$ | 7. $\frac{2}{3}$ | 13. $\frac{1}{3}$ |
| 2. $\frac{3}{5}$ | 8. $\frac{1}{5}$ | 14. $\frac{1}{3}$ |
| 3. $\frac{8}{9}$ | 9. $\frac{4}{5}$ | 15. 0 |
| 4. $\frac{2}{3}$ | 10. $\frac{3}{7}$ | 16. $\frac{1}{3}$ |
| 5. $\frac{1}{3}$ | 11. $\frac{1}{3}$ | 17. $\frac{1}{2}$ |
| 6. $\frac{1}{4}$ | 12. $\frac{1}{6}$ | 18. $\frac{1}{2}$ |
| | | 19. $\frac{1}{6}$ |

20. 1

21. $\frac{1}{3}$

22. $\frac{2}{3}$

23. $\frac{1}{3}$

24. 1

25. $\frac{1}{2}$

26. $\frac{5}{6}$

27. $\frac{1}{6}$

28. $\frac{1}{3}$

Medium (drill)

- | | | |
|-------------------|--------------------|-----------------------------------|
| 1. $\frac{1}{9}$ | 7. $\frac{1}{10}$ | 13. $\frac{4}{5} = 0.8 = 80\%$ |
| 2. $\frac{1}{4}$ | 8. $\frac{1}{10}$ | 14. $\frac{8}{25} = 0.32 = 32\%$ |
| 3. $\frac{1}{2}$ | 9. $\frac{3}{8}$ | 15. $\frac{1}{2} = 0.5 = 50\%$ |
| 4. $\frac{1}{10}$ | 10. $\frac{1}{11}$ | 16. $\frac{19}{20} = 0.95 = 95\%$ |
| 5. $\frac{1}{8}$ | 11. $\frac{1}{10}$ | 17. $\frac{9}{10} = 0.9 = 90\%$ |
| 6. $\frac{1}{11}$ | 12. $\frac{1}{5}$ | 18. $\frac{3}{25} = 0.12 = 12\%$ |

Challenging

- 0.44
- 1.6%
- 0.65

Extension

- $\frac{4}{13}$
- $\frac{3}{13}$

Exit ticket

- Level 1: $\frac{1}{6}$
- Level 2: $\frac{3}{10}$
- Level 3: 30%

11.03 The probability scale (booklet p. 11)

WE DO examples (teacher)

- Ex 1a: B
- Ex 1b: unlikely
- Ex 2a: rolling an odd or even number on a die (C)
- Ex 2b: $\frac{3}{7}$
- Ex 3a: $\frac{3}{14}$
- Ex 3b: 0; 1

Easy (drill)

- | | | |
|---------------|----------------|-----------------|
| 1. likely | 7. likely | 13. even chance |
| 2. likely | 8. unlikely | 14. impossible |
| 3. unlikely | 9. even chance | 15. likely |
| 4. impossible | 10. unlikely | 16. even chance |
| 5. certain | 11. likely | |
| 6. unlikely | 12. unlikely | |

Medium (drill)

- | | | |
|------------------|--------------------|-------------------|
| 1. $\frac{5}{8}$ | 8. 40% | 15. $\frac{1}{3}$ |
| 2. $\frac{2}{5}$ | 9. $\frac{3}{5}$ | 16. 90% |
| 3. 45% | 10. $\frac{7}{10}$ | 17. certain |
| 4. $\frac{3}{8}$ | 11. 65% | 18. unlikely |
| 5. 45% | 12. 80% | 19. unlikely |
| 6. $\frac{2}{5}$ | 13. $\frac{5}{8}$ | 20. likely |
| 7. $\frac{5}{8}$ | 14. $\frac{1}{4}$ | 21. unlikely |
| | | 22. likely |

Challenging

- A at $\frac{1}{2}$, B at $\frac{1}{6}$, C at 1, D at 0

2. own examples

Extension

- 4
- e.g. 2 blue, 7 red, 1 green

Exit ticket

- Level 1: 0
- Level 2: very likely
- Level 3: $\frac{5}{8}$

11.04 Experimental probability (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: $\frac{1}{2}$
- Ex 1b: 40
- Ex 2a: $\frac{3}{10}$
- Ex 2b: fewer heads than expected (24 vs 40)
- Ex 3a: 10
- Ex 3b: the results even out over many trials

Easy (drill)

- | | | |
|---------------------|----------------------|---------------------|
| 1. 23 | 12. $\frac{41}{60}$ | 19. $\frac{7}{50}$ |
| 2. 3 | 13. $\frac{39}{100}$ | 20. $\frac{7}{20}$ |
| 3. 18 | 14. $\frac{7}{30}$ | 21. $\frac{9}{20}$ |
| 4. 4 | 15. $\frac{9}{20}$ | 22. $\frac{23}{60}$ |
| 5. 22 | 16. $\frac{13}{20}$ | 23. $\frac{4}{5}$ |
| 6. 18 | 17. $\frac{37}{50}$ | 24. $\frac{8}{15}$ |
| 7. 26 | 18. $\frac{3}{4}$ | 25. $\frac{2}{3}$ |
| 8. 4 | | |
| 9. 9 | | |
| 10. $\frac{11}{40}$ | | |
| 11. $\frac{19}{30}$ | | |

Medium (drill)

- | | | |
|-----------------------|-------------------------|--------|
| 1. 26% (theory 16.7%) | 5. 30% (theory 16.7%) | 9. 20 |
| 2. 17% (theory 16.7%) | 6. 17.5% (theory 16.7%) | 10. 16 |
| 3. 22% (theory 16.7%) | 7. 16% (theory 16.7%) | 11. 11 |
| 4. 40% (theory 16.7%) | 8. 21% (theory 16.7%) | 12. 27 |
| | | 13. 70 |
| | | 14. 18 |

Challenging

- own results
- answers vary; class results close to $\frac{1}{2}$

Extension

- about 150
- no: red has $P \approx 0.3$, not 0.25

Exit ticket

- Level 1: $\frac{1}{6}$

2. Level 2: 20

3. Level 3: $\frac{5}{24}$

11.05 Complementary events (booklet p. 19)

WE DO examples (teacher)

- Ex 1a: not a queen, $\frac{12}{13}$
- Ex 1b: not a multiple of 3, $\frac{2}{3}$
- Ex 2a: not winning, $\frac{549}{550}$
- Ex 2b: $\frac{1}{5}$
- Ex 3a: $\frac{3}{5}$; $\frac{3}{5}$
- Ex 3b: $\frac{7}{15}$

Easy (drill)

- | | | |
|--------------------|---------------------|----------|
| 1. $\frac{4}{5}$ | 10. $\frac{4}{7}$ | 20. 80% |
| 2. $\frac{3}{4}$ | 11. $\frac{19}{20}$ | 21. 44% |
| 3. $\frac{2}{7}$ | 12. $\frac{6}{7}$ | 22. 0.1 |
| 4. $\frac{5}{6}$ | 13. $\frac{5}{8}$ | 23. 0.98 |
| 5. $\frac{1}{4}$ | 14. $\frac{1}{15}$ | 24. 55% |
| 6. $\frac{1}{6}$ | 15. $\frac{3}{5}$ | 25. 36% |
| 7. $\frac{1}{8}$ | 16. $\frac{1}{12}$ | 26. 0.01 |
| 8. $\frac{11}{12}$ | 17. 0.46 | 27. 57% |
| 9. $\frac{1}{2}$ | 18. 0.56 | 28. 78% |
| | 19. 62% | 29. 0.49 |
| | | 30. 0.44 |
| | | 31. 0.37 |
| | | 32. 41% |

Medium (drill)

- | | | |
|------------------------------------|-------------------------------------|---------------------|
| 1. choosing a consonant | 5. not rolling a 6 (rolling 1 to 5) | 10. $\frac{10}{19}$ |
| 2. not winning (losing or drawing) | 6. tossing tails | 11. $\frac{8}{13}$ |
| 3. it does not rain | 7. failing the test | 12. $\frac{8}{17}$ |
| 4. choosing a black card | 8. rolling 3 or more | 13. $\frac{4}{7}$ |
| | 9. $\frac{7}{10}$ | 14. $\frac{8}{11}$ |

Challenging

- 0.65
- 0.3
- 8

Extension

- $\frac{5}{12}$
- 130

Exit ticket

- Level 1: not rolling a 2
- Level 2: 0.2
- Level 3: $\frac{1}{2}$